Foundational Verification of Probabilistic Programs

Mechanising the Lilac probabilistic separation logic in Lean

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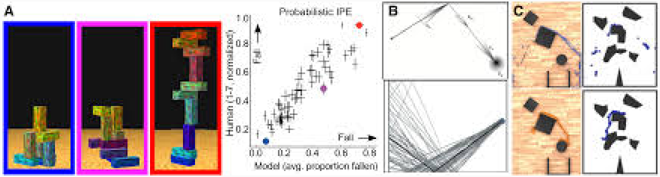
Probabilistic Programming is Everywhere

Probabilistic programming is everywhere

The most general feature of a PPL — sampling from a distribution — lives in the standard library of every major language.

- **Cryptography** — protocols, ZK proofs
- **Differential privacy**
- **Randomised algorithms**
- **Distributed systems**
- **Bayesian inference** & probabilistic ML
- **Scientific simulation** — climate, physics
- **Robotics & perception** — intuitive physics



Discrete vs. continuous

The dividing line that matters runs through the applications.

Discrete	Continuous
Coin flips, Bernoulli / categorical choices, dice. Cryptography, differential privacy, randomised algorithms — traditional CS, well served by *existing* logics.	Gaussian noise, sensor models, Bayesian priors over $\mathbb{R}$, physical quantities. Scientific simulation, Bayesian inference, ML — where the real-world modelling happens.

Continuous distributions are where prior verification work fell short — and where this project makes its **first mechanised** contribution.

Verifying probabilistic programs is notoriously hard

- Conventional testing fails: a rare *correct* outlier execution and a genuine bug look *identical*.
- Correctness is a property of the whole distribution, not of any single run.
- Reasoning about independence and conditioning is subtle and error-prone by hand.

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So we need proof

Machine-checked, **foundational** arguments about distributions — resting on nothing but the axioms of mathematics.

Contribution

The first mechanisation of foundational verification

for probabilistic programs

supporting continuous distributions.

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1. Soundness proofs
2. Interactivity using Iris

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Contribution

using a theorem prover like Lean
or Rocq
Highest achievable assurance
Assuming only the foundational
axioms of mathematics

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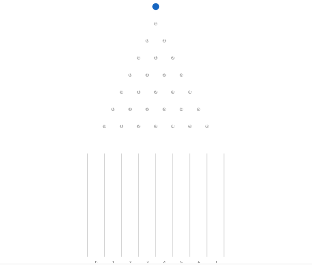
A program is a distribution over runs

$$\text{flip} : (m : \mathbb{R}) \to \text{Measure}\ \{m, m+1\}$$

an individual “run” of the program



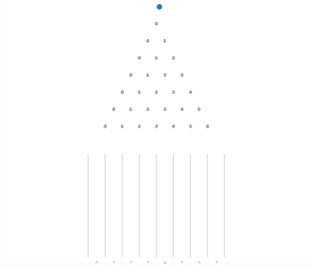
Honing in on a single marble

$$X_0 \leftarrow \text{flip}\ 0$$
$$X_1 \leftarrow \text{flip}\ X_0$$
$$X_2 \leftarrow \text{flip}\ X_1$$
$$\text{ret}\ X_2$$

$$|||$$

$$\text{for}\ (3,\ \text{flip}\ 0,\ \lambda X.\ \text{flip}\ X)$$

The program as a whole



All the marbles at once

Contributions, “revised”

Now that the motivation is in place: what did we mechanise, and why does it matter?

- Lilac is a probabilistic separation logic whose guiding idea is

  $$\text{“}\textit{separation} = \textit{independence of random variables}\text{”}$$

  The separating conjunction $P * Q$ asserts that the random variables described by P and Q are probabilistically independent — and Lilac supports **continuous** distributions.
- We give the **first mechanisation of Core[1] Lilac in Lean 4**, the first mechanisation of *any* probabilistic separation logic in Lean.

[1]Extended Lilac adds reasoning about conditioning

What was mechanised

1. **Soundness, from the ground up.** Core Lilac is built as a model over probability spaces with finite footprint, and every program-logic proof rule is proved sound against that model:

`wp_ret` `wp_bind` `wp_unif` `wp_cons` `wp_frame`

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2. **An interactive proof interface.** Lilac is instantiated as a bunched-implication logic on the Lean port of Iris, so proofs are carried out in the *Iris proof mode*.
3. **Worked examples.** Specifications of sampling programs proved in that proof mode — including constructing a separating conjunct of two independent uniform samples.

Separation *is* independence — formally

The separating conjunction splits a probability space into an independent product $\sigma_1 \star \sigma_2$:

```
sep P Q := ⟨fun σ ↦ ∃ σ₁ σ₂, ∃ (σ₁₂ : ✓'(σ₁ ⋆ σ₂)),
                          ↓σ₁₂ ≤ σ ∧ P.1 σ₁ ∧ Q.1 σ₂, …⟩
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This is not just a slogan — it is a theorem. Owning two random variables separately entails that they are genuinely independent:

```
lemma indep_of_own {X Y : RV ⟪Ty.real⟫} (Ω : PSp)
    (ownXY : (iprop(own X ∗ own Y) : LProp).1 Ω) :
    IndepFun (μ := Ω.μ) X Y
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$$(\text{own } X * \text{own } Y) \implies X \perp\!\!\!\perp Y$$

with the magic wand $-\!\!*$ available for hypothetical reasoning.